

Implementation Summary: CampusNote Pro SaaS Prototype

Project: CampusNote Pro

Milestone: HW5 - First SaaS Prototype

Date: May 6, 2026

1. What is Implemented

The current prototype establishes a functional "vertical slice" encompassing the core lifecycle of a document. The following requirements from the functional specification have been realized:

- Identity & Access Management: Basic authentication logic via AuthController, credential validation, and session termination (FR-ST-01, 02, 03, 05).
- Document Ingestion: PDF upload capability and "Draft" state management via DocumentService (FR-ST-15, 16, 20, 21).
- AI Ranking (Simulated): Implementation of an internal AIRankingService that generates quality scores, currently used during manual administrative reviews (FR-ST-23, 24, 27, 28).
- Gamification Logic: Calculation of CampusCoin rewards during document approval and persistence in the user's virtual ledger (FR-ST-33, 34).
- Search & Retrieval: Basic keyword-based searching restricted to "Published" content and file download tracking (FR-ST-39, 40, 44).
- User Analytics (Partial): Display of personal upload history and download/view counts (FR-ST-11, 12).

2. What Works

The prototype demonstrates a functional flow from the presentation layer to the data layer:

- Document Pipeline: A user can select a PDF file via the UI, which is successfully transmitted to the backend and saved as a DRAFT status in the database.
- Administrative Review: The system allows an administrator to review a draft, assign a score, and transition the status to PUBLISHED, which triggers a coin reward for the uploader.
- Cloud Deployment Ready: The application is containerized via Docker and configured for hosting on platforms like Heroku.
- Persistence Integrity: All state changes (status updates and coin balances) are persisted in the PostgreSQL instance.

3. What is Not Yet Implemented

To maintain the focus on the "vertical slice" and ensure honesty regarding current progress, the following features remain in the development backlog:

- Automated Testing Suite: Full implementation of Unit, Functional, and BDD Acceptance tests (Given/When/Then) is scheduled for the next iteration.

- **Automated AI Triggering:** The system does not yet automatically trigger evaluation upon upload; it currently relies on the administrative review step.
- **Stateless Authentication (JWT):** The current prototype uses HttpSession-based authentication; migrating to JWT is planned for the next milestone.
- **Advanced AI Extraction:** In-depth text extraction from PDFs and automated rejection routing (FR-ST-25, 26, 29, 30).
- **Social Engagement:** Global leaderboards and document liking features are structurally in place but require further UI polish (FR-ST-31, 32, 35, 36, 37, 38).

4. Key Design Decisions

The development process followed these architectural choices:

- **3-Tier Architecture:** Maintaining clear boundaries between the Presentation Layer (React), the Logic Layer (DocumentService, GamificationService), and the Data Layer (PostgreSQL).
- **RESTful Service Design:** The backend exposes endpoints using standard HTTP methods via specialized Controllers (DocumentController, AuthController, etc.).
- **Modular Monolith:** Logic is organized into distinct packages (CoreSecurity, CoreDocumentManagement, CoreGamification) to ensure separation of concerns.
- **Session-Based Security:** Adopting standard Spring Session management for the initial prototype to ensure a stable and demonstrable login flow.